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CS 555: Homework # 2
Spring 2026
GitHub Repository: `https://github.com/redhorn144/CS555`

Problem 1
optional: summarize the problem

Solution:

(a) For this part we have implemented a simple the point finite difference stencil to solve Burgers' equation. Plots of the solution for both Chebyshev and uniform nodes are shown in Figures 1 and 2, respectively. For both of these N is taken to be 200, and the time step is taken to be smaller than the CFL constrint from part c. Stapshots of the solution are taken at for $t = 0.1k, \quad 0 \leq k \leq 20$. In these particular cases we can see that the Chebyshev nodes do a better job at resolving the shock in the boundary layer. This is because the Chebyshev nodes are clustered more closely together near the boundaries. To be consistent with the paper we take $\nu = 1/100\pi$, but to be consistent with the homework handout we take $u^0 = \sin(\pi x)$ instead of $u^0 = -\sin(\pi x)$.

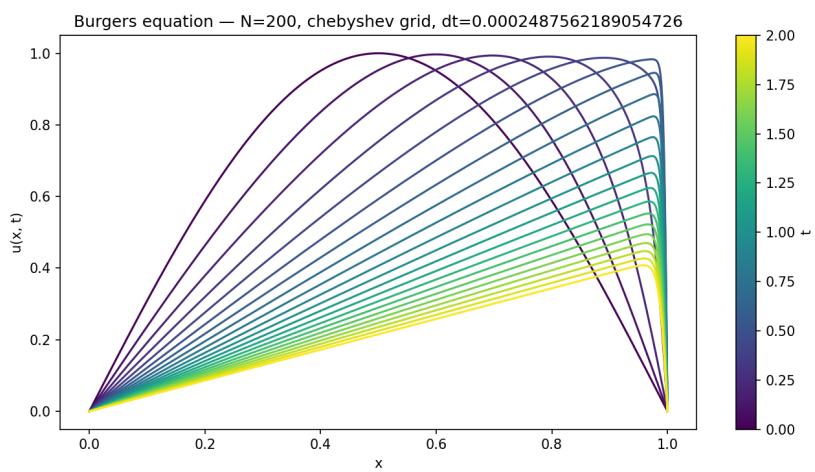

Figure 1: Solutions to Burgers' equation with Chebyshev nodes.

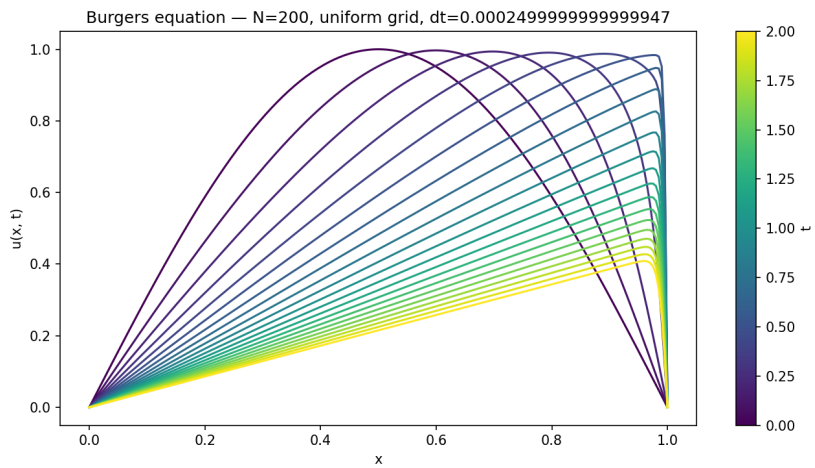

Figure 2: Solutions to Burgers' equation with uniform nodes.

(b) Now we are asked to calculate the maximum value of the derivative of u $s(t) = max_x|u_x(x,t)|$. This is done with N = 200 again for both the uniform nodes and the Chebyshev nodes. The plot of $s(t)$ as well as $s^* = max_t s(t)$ is shown in Figures 3 and 4 for the Chebyshev and uniform nodes, respectively. The paper scales $t* = argmax s(t)$ by π. But without that scaling the analytical values of s^* and t^* are $s^* = 152.00516$ and $t^* = 0.51047$. The Chebyshev node case does notably better for determining s^* and t^*.

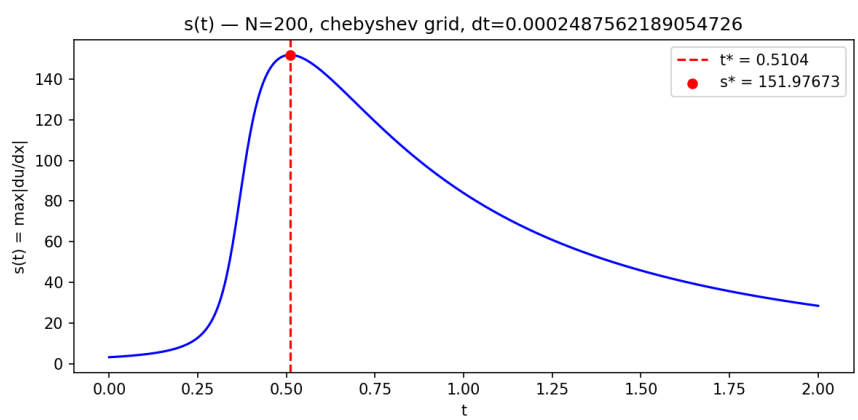

Figure 3: Your descriptive caption here.

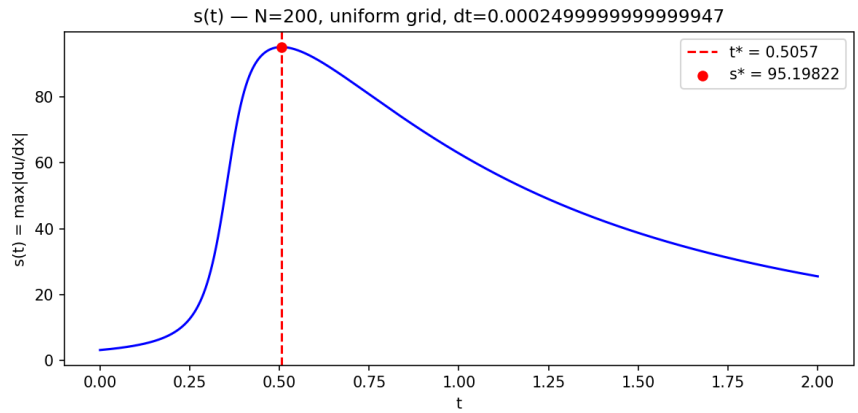

Figure 4: Your descriptive caption here.

(c) Now, we push N with Chebyshev nodes up to 15000 and observe convergence of s^* to the analytical value of 152.00516. We see, with that by N = 10000 we have a relative error of $4.552e-08$ which is on the order of the rounding error of the analytical value. Because the table has s^* out to 5 decimal places, as the analytical value is 152.00516, we match by N = 10000. This notably outperforms the uniform node case substantially.

N	dt	nsteps	s^*	rel_err
25	3.864938e-03	518	72.28430	5.245e-01
50	1.960784e-03	1020	142.29624	6.387e-02
100	9.900990e-04	2020	151.39811	3.994e-03
200	4.975124e-04	4021	151.97705	1.849e-04
400	2.493766e-04	8021	152.00590	4.878e-06
800	1.248439e-04	16020	152.00583	4.419e-06
1500	6.662225e-05	30020	152.00538	1.481e-06
3000	3.332223e-05	60020	152.00522	3.941e-07
6000	1.666389e-05	120020	152.00517	1.073e-07
10000	9.999000e-06	200020	152.00516	4.552e-08
15000	6.666222e-06	300020	152.00516	2.601e-08

Table 1: Convergence data for various N.

(d) In this part we are asked to measure s^* for the conservation form of Burgers' equation. The plot of $s(t)$ for the conservation form is shown in Figure 5. We are asked to compare this at the highest resolution to the convective form, and we see that they are very close to each other. Infact they both predict $s^* = 152.00516$ to 5 decimal places, which is the same as the analytical value. We see that the conservation form also converges to the analytical value as we increase N. Part (e) elucidates the convergence behavior of both the convective and conservation forms for better comparison.

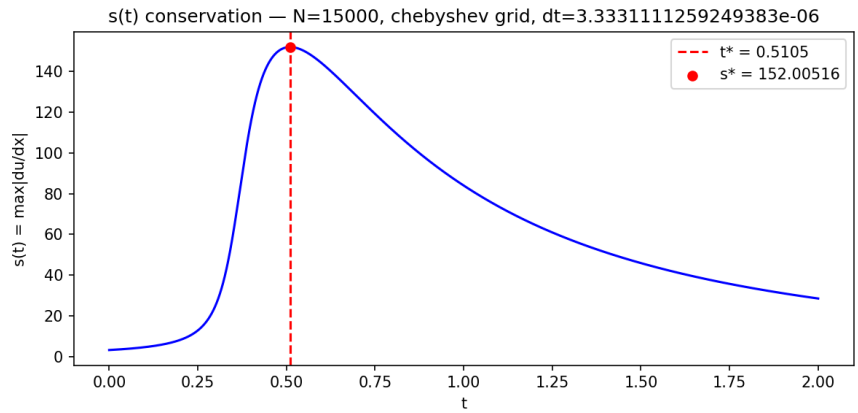

Figure 5: Conservation of $s_{\max}$ using Chebyshev nodes.

(e)

The following tables document the convergence behavior for $N = 2^k$ $(k = 5, \ldots, 11)$ across four combinations of spatial discretization and nonlinear formulation. As instructed, the timestep $\Delta t = 0.1/N$, derived from the uniform grid CFL condition, is held constant across all cases to ensure a direct comparison without introducing variable time-stepping artifacts. The convergence ratio is defined as $\mathrm{err}_{N/2}/\mathrm{err}_N$.

Table 2: Case i. Uniform spacing, convective form

N	nsteps	s^*	rel.err.	ratio
32	640	204.75163	3.47004×10^{-1}	-
64	1280	264.95618	7.43074×10^{-1}	0.467
128	2560	241.56922	5.89217×10^{-1}	1.261
256	5120	192.58263	2.66948×10^{-1}	2.207
512	10240	164.69320	8.34711×10^{-2}	3.198
1024	20480	155.38910	2.22620×10^{-2}	3.749
2048	40960	152.86551	5.66003×10^{-3}	3.933

Table 3: Case ii. Chebyshev spacing, convective form

N	nsteps	s^*	rel.err.	ratio
32	640	167.45688	1.01653×10^{-1}	-
64	1280	153.42089	9.31369×10^{-3}	10.914
128	2560	152.11972	7.53643×10^{-4}	12.358
256	5120	152.01843	8.72962×10^{-5}	8.633
512	10240	152.00751	1.54404×10^{-5}	5.654
1024	20480	152.00569	3.48080×10^{-6}	4.436
2048	40960	152.00529	8.50164×10^{-7}	4.094

Table 4: Case iii. Uniform spacing, conservation form

N	nsteps	s^*	rel.err.	ratio
32	640	65.52165	5.68951×10^{-1}	-
64	1280	102.90704	3.23003×10^{-1}	1.761
128	2560	140.71434	7.42792×10^{-2}	4.348
256	5120	157.73907	3.77218×10^{-2}	1.969
512	10240	156.53456	2.97977×10^{-2}	1.266
1024	20480	153.55362	1.01869×10^{-2}	2.925
2048	40960	152.42586	2.76767×10^{-3}	3.681

Table 5: Case iv. Chebyshev spacing, conservation form

N	nsteps	s^*	rel.err.	ratio
32	640	159.19227	4.72820×10^{-2}	-
64	1280	153.81930	1.19348×10^{-2}	3.962
128	2560	152.15108	9.59988×10^{-4}	12.432
256	5120	152.01876	8.94915×10^{-5}	10.727
512	10240	152.00711	1.28268×10^{-5}	6.977
1024	20480	152.00556	2.62221×10^{-6}	4.892
2048	40960	152.00525	6.23135×10^{-7}	4.208

Discussion

- **Grid Spacing (Uniform vs. Chebyshev):** The uniform grid severely under-resolves the sharp internal layer at lower N values, leading to significant numerical dispersion. This manifests as poor approximations of s^* and erratic error convergence until $N \geq 1024$. In contrast, Chebyshev spacing naturally clusters grid points at the boundaries, placing high resolution precisely where the local gradients are maximized. Consequently, the Chebyshev cases reach an $\approx 10^{-5}$ relative error by just $N = 256$, outperforming the uniform grid's finest resolution.

- **Convergence Ratio:** The theoretical error ratio for a second-order scheme when halving the spatial step size is ≈ 4. For the Chebyshev grids (Cases ii and iv), the ratio rapidly stabilizes and approaches ≈ 4.1 by the finest resolution, strictly confirming the scheme's second-order spatial accuracy. For uniform grids, the ratio behaves erratically before finally approaching the theoretical asymptotic value once the grid properly resolves the boundary layer at high N.

- **Formulation (Convective vs. Conservation):** At lower resolutions, the discrete truncation errors of the two forms lead to distinct failure modes; the convective form overestimates s^* due to spurious oscillations, while the conservation form under-predicts it. However, once the grid is sufficiently refined (especially with Chebyshev spacing), both formulations converge to the analytical maximum $s^* \approx 152.005$ with nearly identical precision, robustly validating their formal equivalence.